2026/2027	**Forsyth County Schools** **High School (Non-EOC) Course Syllabus** **Cloud Computing — Denmark High School**

Teacher Information

Teacher Name: **Kinny Cornelison**

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Canvas is the primary channel for assignments, due dates, and course announcements. Email is the fastest way to reach me directly; I respond within one school day.

Availability for Extra Help:

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By appointment only. Email me at least one school day in advance and we will set a time before school, after school, or during my planning period.

Materials Needed:

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- Charged school-issued device, every day — this is a daily-use requirement, not an occasional one
- Forsyth County Google Workspace account (Drive, Sheets, NotebookLM)
- CodeHS account (provided by the school — no cost to families)
- AWS Academy account (provided by the school — no cost to families)
- Wired earbuds or headphones
- Composition or spiral notebook for network diagrams, command reference, and notes by hand
- Pen or pencil

No textbook purchase is required for this course.

Course Information

Course Description:	Cloud Computing is the third and capstone course in Denmark High School's Information Technology — Cloud Computing pathway. Its purpose is to take students from "the cloud is somewhere else" to "I know what is running, where it lives, what it costs, and what breaks." The first semester is built from the hardware up. Students study what a cloud actually is and why organizations move to one, then go underneath the abstraction: processors and accelerators, operating systems and the command line, and computer networking through the OSI model and IPv4 addressing. The emphasis is on being able to reason through an unfamiliar problem rather than recall a definition. The second semester turns that foundation toward certification and application. Students cover storage and redundancy, disaster recovery, identity and access management, the shared responsibility security model, architecture and scaling, and cloud cost management — building real resources in a live cloud console throughout. Students prepare for and sit two industry certification examinations: AWS Certified Cloud Practitioner and Microsoft Azure Fundamentals.

	Students work in CodeHS, AWS Academy, and a classroom hardware lab. This course is hands-on and fast-paced, and it is designed so that students leave able to build, troubleshoot, and explain — not only to pass an exam.
Course Standards:	Georgia Department of Education — Career, Technical and Agricultural Education (CTAE), Information Technology Career Cluster, Cloud Computing pathway. Course standards: Cloud Computing. The complete standards list is posted in Canvas.
Course Text(s) and Resources:	No textbook is required. The course runs on the following platforms and resources: • CodeHS — instruction, practice, and auto-graded exercises for computing systems, networking, and security fundamentals • AWS Academy Cloud Foundations — hands-on cloud labs and certification-aligned curriculum • Microsoft Learn — Azure Fundamentals learning paths and courseware • AWS Skill Builder — official examination preparation and practice questions • Classroom hardware lab — servers, networking equipment, and storage for on-premises comparison work • NotebookLM — unit-level AI reference notebooks built from course materials, available to students at any time • Canvas — assignments, due dates, and the most current course information

Projected Course Timeline

Unit	Approximate Timeline	Major Assignments
Semester 1		
1 — Cloud Foundations and the Business Case	Weeks 1–3 (August)	• Course setup: CodeHS, AWS Academy, NotebookLM, and career portfolio onboarding • Service model comparison (IaaS / PaaS / SaaS) with scenario justification • On-premises vs. cloud cost-and-capacity brief • Unit 1 vocabulary and concepts assessment
2 — Inside the Machine: Compute, Memory, and Accelerators	Weeks 4–6 (August–September)	• CPU / GPU / NPU workload-matching lab • x86 vs. ARM architecture comparison brief • Instance-state diagram and billing analysis (pending through terminated) • Unit 2 assessment
3 — Operating Systems and the Command Line	Weeks 7–11 (September–October)	• Linux file system and permissions lab • Paired bash scripting set (each partner must be able to explain every line) • SSH, remote session, and web server configuration lab • Unit 3 practical: complete a stated task entirely from the command line
4 — Networking: OSI, IPv4, and Cloud Networks	Weeks 12–17 (October–December)	• OSI layer identification and troubleshooting drills • IPv4 addressing, CIDR, and subnet design lab • DNS, NAT, and port behavior investigation using command-line tools

		• Virtual private network build using an infrastructure-as-code template • Semester 1 exam
Semester 2		
5 — Storage, Redundancy, and Disaster Recovery	Weeks 18–21 (January)	• RAID levels 0 through 6 (and 60) redundancy analysis • Object, block, and file storage selection lab; create and attach a storage volume • Object storage bucket build with a static site. Upload to career portfolio • Disaster recovery plan with stated RTO and RPO targets
6 — Security, Identity, and Shared Responsibility	Weeks 22–26 (January–February)	• Shared responsibility model analysis across IaaS, PaaS, and SaaS • Identity and access management lab: users, groups, roles, and policies • Application architecture study: where authentication happens and what it costs you • Security audit: find the lapses in a provided configuration and recommend fixes
7 — Architecture, Scaling, and Cost Management	Weeks 27–31 (February–March)	• Load balancer configuration and performance evaluation • Auto scaling group with launch template and monitoring plan. Upload to career portfolio • Serverless compute function build. Upload to career portfolio • Total cost of ownership analysis using provider pricing calculators • AWS Certified Cloud Practitioner examination window (weeks 30–31)
8 — Emerging Technology, Certification, and Capstone	Weeks 32–36 (April–May). Weeks 37–38 are reserved for end-of-course and final testing.	• Machine learning, IoT, big data, and blockchain use-case briefs • Infrastructure-as-code template addressing a stated user need. Upload to career portfolio • Microsoft Azure Fundamentals examination window (weeks 35–36) • Capstone: design, build, document, and defend a complete cloud solution • Career portfolio final review • Semester 2 exam

Access this course in Canvas for the most up-to-date course information, including ancillary materials and key dates.

Assessment & Grading

Assessment Categories		
• **Formative Assessments** include, but are not limited to, homework, class work, labs, investigations, practice tests, rough drafts, and sections of projects/research papers/presentations. • **Summative Assessments** include, but are not limited to, unit tests, final projects, final essays, formal lab reports, final research papers, and final presentations.		
Grading Scale	**Grading Weights**	**Grading Calculations**

A = 90-100 **B = 80-89** **C = 70-79** **F = 0-69**	Formative = 25% + Summative = 75% *(midterm and final exam multiplier=1)*	50% (Semester 1) + 50% (Semester 2) = Course Average

Final Exam Exemption: Students may exempt the final exam if they have a 90 or above course average.

Make-up Work

Make-up work is defined as work assigned during a student's absence, not work assigned prior to an absence. Students who are absent from school have five (5) school days upon returning to school to complete makeup work. The teacher has discretion to grant a longer period of time to make up the work if there are extenuating circumstances.

Summative Reassessment

To support continued learning, students may engage in one summative reassessment opportunity per semester per course.

Procedures:

- All formative assessments must be complete before a student takes a corresponding summative reassessment.
- The student must engage in re-teaching activities prior to the summative reassessment.
- The teacher collaborative team can determine one of two methods for reassessment:
  - Engage the student in re-teaching activities and reassess the standards.
  - Engage the student in re-teaching activities and use the midterm or final exam to reassess the aligned standards.
- To ensure accuracy in grading and reporting practices, full credit is applied to all reassessments.
- When a student does not show minimal proficiency of standards on multiple summative assessments, a multi-tiered system of supports will be put in place to assist the student in reaching proficiency.

Course-Specific Reassessment Procedures — Cloud Computing

- Students may reassess one (1) summative TEST per semester. Projects, capstones, hands-on practicals, and performance tasks are not eligible for reassessment.
- The reassessment window opens near the end of each semester. This is deliberate: it lets each student look back across the whole semester, identify their lowest summative test grade, and target that one.
- Before reassessing, the student must complete all corresponding formative work and engage in re-teaching — the relevant CodeHS review set, the relevant AWS Academy or Microsoft Learn module, and a conference with me.
- The reassessment will be a different version of the assessment covering the same standards.
- Full credit applies to reassessments taken under this policy. A retake required as a consequence of an academic integrity violation is scored under the Academic Integrity policy below and does not consume the one reassessment allowed per semester.

Student Responsibilities

Academic Integrity

It is the expectation of Forsyth County Schools that students exhibit academic integrity when submitting evidence of learning. When students submit evidence that does not authentically represent what they know and can do, the evidence is rendered invalid. Examples of academic dishonesty are outlined in both the Family Engagement Guide & Student Handbook and The Forsyth County Schools Code of Conduct. Depending on the severity of the situation, academic dishonesty/cheating may result in the loss of credit, completing an alternative assignment and/or disciplinary consequences up to a tribunal referral.

Consequences for High School Students:

- First Offense: Parent contact + academic integrity contract + take a different assessment with a maximum assessment score of up to 60%
- Second Offense: Parent contact + loss of credit up to 100% + removal from honor clubs/societies + possible in-school suspension or out-of-school suspension
- Third Offense: Parent conference + loss of credit up to 100% + possible out-of-school suspension

- Academic Integrity offenses will not reset each school year. For example: If a student has an academic integrity offense during 9th grade and another one in 10th grade, the offense in 10th grade would be considered their second overall offense.
- Standardized Assessments: Any suspicion or instances of cheating on standardized tests will be reported to the governing agency and, based on the severity of the behavior, will receive disciplinary consequences.
- Severe Offenses: Should a student engage in behavior that compromises the integrity of an assessment (i.e., theft of assessment, pictures/dissemination of content, etc.), immediate and more severe consequences (up to a disciplinary tribunal) may be applied.

**Students who face consequences for submitting work that does not reflect their own knowledge and skills may only reassess in accordance with the FCS Academic Integrity Consequences for High School Students outlined above.*

Course-Specific Clarification — Cloud Computing

- A student who must take a different assessment as a consequence of an academic integrity violation will have that grade calculated as: score earned × 0.60.
- Examples: a 100 on the replacement assessment is recorded as a 60. A 90 is recorded as a 54. An 80 is recorded as a 48.
- The maximum grade that can be recorded for an academic integrity retake is therefore a 60. This formula is applied the same way every time so there is no question about how the grade was determined.
- An academic integrity retake is separate from the one summative reassessment allowed per semester. It does not count as that reassessment, and it does not use it up.

Responsible Use of Artificial Intelligence

Artificial Intelligence refers to computer systems that are trained with data to accomplish tasks normally requiring human intelligence. AI is used across industries and communities and can serve as a helpful tool in educational settings. Students should learn how to evaluate if and when they should use AI and develop responsible and ethical practices when using AI tools. Students with Forsyth County Schools are expected to uphold the highest standards of academic integrity. This includes the responsible use of emerging technologies such as artificial intelligence.

Students will:

- Use approved age-appropriate AI tools for educational purposes only
- Follow teacher guidance on the use of AI tools
- Submit work that is fundamentally my own and provide proper citations when AI is used
- Evaluate the purpose, safety, and the trustworthiness of AI tools before use
- Review and critically assess output of AI tools recognizing that AI may contain bias and inaccuracies
- Not use AI tools to propagate harmful, misleading, offensive or inappropriate content

All members of Forsyth County Schools' community agree to follow the **Forsyth County Schools Code of Conduct**, school rules and commit to these responsible use guidelines.

Course-Specific AI Guidance — Cloud Computing

This course uses AI tools deliberately, as part of instruction. Learning to work alongside AI is one of the skills being taught. For every task, students will be told clearly whether AI assistance is permitted, required, or prohibited.

- AI-permitted tasks: students may use an approved AI tool to research services, explain error messages, or draft configuration and scripts, but must be able to explain, trace, and modify every line and every setting they submit. "The AI wrote it" is never an acceptable explanation.
- AI-prohibited tasks: hands-on practicals, command-line assessments, network design and troubleshooting assessments, and all certification preparation testing are completed without AI assistance. Using AI on these is an academic integrity violation.
- Submitting AI-generated work a student cannot explain does not represent that student's learning, and will be treated as invalid evidence of learning.
- Any use of AI beyond what is expressly stated as permitted for a given assignment is a violation of the Forsyth County Schools academic integrity policy and will be treated as academic dishonesty. If an assignment does not say AI is allowed, assume it is not. Ask me before you use it — asking is always free; assuming is not.
- This applies to every AI tool, not only the ones used in class, and to work completed outside of school as well as in the

classroom.

School/Content Team Information

Denmark High School — Home of the Danes
645 Mullinax Road, Alpharetta, GA 30004
Main office: 470-533-2521 | Website: dhs.forsyth.k12.ga.us | CEEB code: 111000
Counseling secretary: 470-533-2521, ext. 511401
Department: Career, Technical and Agricultural Education (CTAE) — Information Technology and Cloud Computing pathway
Principal: Kim Oliver
Assistant Principal over CTAE: Glenda Santiago
CTAE Department Head: Eric Arnold
Collaborative content team: Kinny Cornelison (sole instructor for this course)

Syllabus Acknowledgment

By signing below, the student and parent/guardian acknowledge that they have read and understand this syllabus, including the course expectations, grading and reassessment policies, the academic integrity policy and its grade calculation, and the guidelines for AI use in this course.

This form can be completed and signed on a device using Adobe Acrobat Reader (free). Click a signature box and follow the prompts — Acrobat will walk you through creating a digital ID the first time. You may also print the page and sign it by hand. Return the completed page to Mr. Cornelison. Questions are welcome at any time.

Student Name (printed):

Student Signature:

Date Signed:

Parent/Guardian Name (printed):

Parent/Guardian Signature:

Date Signed:

Best number to reach a parent/guardian:

Parent/Guardian email: